

LLM Perception of Time

A Six-Experiment Scoping Study — Methods, Results, and Reproduction

Time Project (sandbox exploration)

Compiled June 11, 2026

Repository: `explorations/time/` Branch: `claude/time-project-experiments-g0k16r`

Abstract

We ask whether large language models perceive time. The question conflates three distinct problems, and we separate them with six controlled, API-only experiments (no training). **Self-perception (E1, E2, E5, E6):** a model’s apparent ability to estimate its own generation latency is *entirely a proxy for its own output length*. It is well calibrated when latency is output-driven (Spearman $\rho = 0.79$) and collapses to zero when latency is driven by hidden reasoning or input prefill ($\rho \approx 0$); estimating in token space beats seconds ($\rho 0.89$ vs 0.66); and the systematic $\sim 2\times$ length undershoot is mostly fixable in-context (geometric-mean ratio $0.37 \rightarrow 0.76$) for non-reasoning models. **Agency (E3, E4):** agents act on elapsed time only when it is explicit; an agent’s correct time-sensitive decision rate rises from 0.49 (no signal) to 0.80 (text timestamps) to 0.96 (an explicit harness-injected elapsed-time field). The unifying finding: **LLMs have no internal clock — they have a length estimator, and they respond to elapsed time only when it is handed to them.** The single recurring boundary across every experiment is the *reasoning model*, whose latency hides in tokens it cannot see and which no in-context fix repairs. This document is self-contained and reproducible: every number below is regenerated by the committed scripts from the committed raw data.

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1 Scope and the three lanes

“Does an LLM perceive time?” is really three questions that are mechanistically unrelated. A model can be strong at one and hopeless at another:

1. **Temporal reasoning in text** — ordering events, date arithmetic, durations stated in language. Well studied and largely saturated (TimeBench, TRAM, Test-of-Time). We do not pursue it.
2. **Temporal self-perception** — can a model estimate the time/length of its *own* computation? Experiments E1, E2, E5, E6.
3. **Time-aware agency** — does an agent change behaviour based on elapsed time in a transcript? Experiments E3, E4.

The motivation, literature notes and reading list are in `research-notes.md`. The experiment plan is in `experiments.md`; the cross-experiment synthesis in `SUMMARY.md`; and an alternative “train-it-in” research direction in `direction-train-it-in.md`. This report consolidates all of it.

2 Shared infrastructure and reproducibility

2.1 Models

All experiments draw from one fixed roster, defined in `common.py`. Exact API model IDs as used (the study was run on 2026-06-11):

Roster key	Provider	API model ID	Reasoning model?
haiku	Anthropic	claude-haiku-4-5-20251001	no
sonnet	Anthropic	claude-sonnet-4-6	no
opus	Anthropic	claude-opus-4-8	no
gpt4o-mini	OpenAI	gpt-4o-mini	no
gpt4o	OpenAI	gpt-4o	no
gpt5	OpenAI	gpt-5	yes
gpt5.2	OpenAI	gpt-5.2	yes
o4-mini	OpenAI	o4-mini	yes

Reasoning models spend hidden reasoning tokens, so their wall-clock latency decouples from visible output. `common.py` handles the two provider quirks discovered empirically: OpenAI reasoning models require `max_completion_tokens` (not `max_tokens`) and reject a `temperature` argument; `claude-opus-4-8` also rejects `temperature` in this harness (so E4 passes `temperature=None` for opus). Reasoning models have their token budget bumped to ≥ 4096 because hidden reasoning consumes the output budget.

2.2 The call interface

Every experiment imports one helper so provider logic is defined once:

```
from common import call # roster key -> unified call
r = call("sonnet", prompt, max_tokens=1024)
# r: Result(model, text, latency_s, input_tokens, output_tokens, ok, error)
```

`latency_s` is wall-clock measured with `time.perf_counter()` around the network call; it includes network and queueing time, not only generation. Retries use exponential backoff (4 attempts).

2.3 Environment

- Python 3.11; virtualenv at `explorations/time/.venv`. Pinned dependencies in `requirements.txt` (anthropic 0.109.1, openai 2.41.1, numpy 2.4.6, pandas 3.0.3, scipy 1.17.1, matplotlib 3.10.9).
- Secrets via environment variables `ANTHROPIC_API_KEY` and `OPENAI_API_KEY`.

2.4 Data format and the resume/dedup scheme

Each experiment directory `eN-*/` contains: `run.py` (collect raw data), `analyze.py` (compute statistics and figures), `results.jsonl` (one JSON object per API call), `REPORT.md` (self-contained findings), and PNG figures. Collection is *append-only*; `run.py` is idempotent via a `load_done()` check on a logical key (typically `(model, task/scenario, trial, condition)`), so an interrupted run resumes without recomputation.

Known hazard (documented, not silent). This `append + load_done()` scheme is **not safe against concurrent writers**: two `run.py` processes for the same experiment will both pass `load_done()` and write duplicate rows. This actually occurred during the run (see §11). The fix is `dedup.py`, which canonicalizes each `results.jsonl` by its logical key (keeping the last occurrence) before analysis. `analyze.py` scripts also tolerate duplicates, but `dedup.py` guarantees exact counts.

2.5 How to reproduce

From `explorations/time/`:

```
python -m venv .venv && . .venv/bin/activate
pip install -r requirements.txt
export ANTHROPIC_API_KEY=... OPENAI_API_KEY=...

# Re-derive every table and figure from the committed raw data (no API calls):
for d in e1-self-duration e2-token-time e3-log-gap-probe \
    e4-harness-clock e5-latency-decoupled e6-length-bias; do
    ( cd $d && python analyze.py )
done
```

```
# To re-collect from scratch (spends API budget), per experiment:
# cd eN-* && python run.py # resumable; dedups via ../dedup.py if needed
```

Re-running `analyze.py` on the committed `results.jsonl` reproduces every statistic in this report exactly. Re-collecting with `run.py` reproduces them up to sampling noise.

2.6 Determinism caveats

Calls use provider-default temperature (not pinned), so re-collection varies trial-to-trial. Absolute wall-clock numbers are harness- and hardware-specific. We therefore lead with *rank* statistics (Spearman ρ , ordering accuracy) and *between-condition* contrasts, which are robust, and treat absolute latencies as distributions (multiple trials, reported coefficient of variation) rather than points.

3 E1 — Self-duration calibration

Question. Can a model estimate its own wall-clock generation time? **Hypothesis (from the literature).** No: estimates overshoot by 4–7 $\times$ and ordering is at chance.

Setup. 12 tasks spanning short→long outputs. For each (model $\times$ task $\times$ 3 trials): a PRE estimate (“how many seconds will your response take?”), the real generation (record `latency_s`, `output_tokens`), and a POST estimate (“how long did that take?”). 6 models, 648 calls, 216 units.

Results (Table 1, Fig. 1). The hypothesis is *rejected* for this task set. Estimate/actual geometric-mean ratios sit near 1 (overall 0.81), not 4–7 $\times$; ordering accuracy is 0.79–0.95 (chance = 0.5); Spearman ρ_{PRE} is 0.74–0.94, all $p < 0.001$. Trial-to-trial latency noise is small (median CV 0.04–0.14), so the calibration signal is real.

Table 1: E1: per-model calibration. Ratio is geometric-mean estimate/actual.

model	gm ratio PRE	ρ_{PRE}	ordering acc.	ρ_{POST}
opus	0.93	0.94	0.94	0.95
sonnet	0.76	0.85	0.95	0.85
gpt4o	1.04	0.83	0.86	0.77
gpt4o-mini	1.36	0.74	0.86	0.89
haiku	0.86	0.89	0.88	0.76
gpt5.2	0.32	0.74	0.79	0.84
overall	0.81	0.75	—	—

Why the contradiction. The task set varies tasks primarily by *output length*, and for non-reasoning models latency is output-token-dominated. Models can anticipate roughly how long their own output will be, so they predict latency well via an implicit length proxy — not via a clock. Two tells: gpt5.2 (reasoning) is the worst calibrated (0.32 $\times$, it undershoots because hidden reasoning latency is invisible), and undershoots cluster on free-form generations (`code`, `story`). E5 (§7) confirms this mechanism and reconciles the result with the literature.

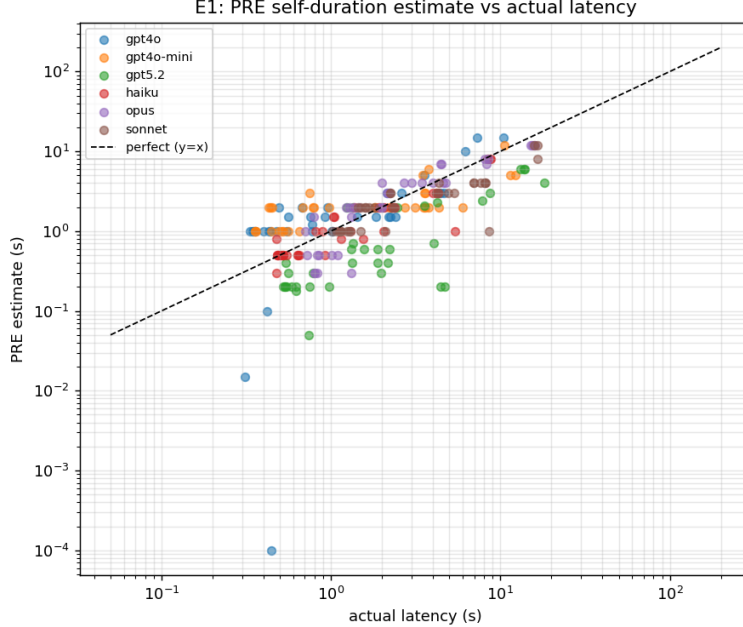


Figure 1: E1: estimated vs. actual latency (log-log), pooled. Tight diagonal structure — models rank and roughly size their own latency.

Verdict. Rejected *for this regime*: models predict their own latency well when it is output-driven (ρ up to 0.94), via a length proxy, not a clock.

4 E2 — Token/step proxy vs second-space

Hypothesis. Output length is something the model has more access to than time, so estimating in tokens should be better-calibrated than in seconds.

Setup. Same task family. Four conditions per (model $\times$ task $\times$ 3): (a) predict seconds; (b) predict output tokens; (c) predict tokens then convert via a measured per-model tokens/sec constant; (d) reason about size first, then predict seconds. 6 models, 864 calls.

Results (Table 2, Fig. 2). Token-space beats second-space: pooled ρ 0.889 vs 0.657 ($\Delta = +0.231$). Per-model, seconds estimates correlate well *within* a model but the scale is wildly model-specific (gpt4o-mini overshoots $6.6\times$, o4-mini undershoots $0.5\times$), which destroys the pooled correlation; tokens normalize this. Models under-predict their own length $\sim 2\times$ (token ratio 0.49). Converting tokens to seconds (c) compounds the undershoot (0.24). Reasoning first (d) does *not* help (0.545).

Verdict. Supported. Models track their own output length, not time; the practical target for any self-estimation is token/step space plus a bias correction for the $\sim 2\times$ undershoot. The approach fails for reasoning models (o4-mini token ratio 0.10), the recurring boundary.

Table 2: E2: pooled calibration by estimation condition.

condition	ρ (pred vs actual)	gm(pred/actual)
(a) seconds	0.657	2.16
(b) tokens	0.889	0.49
(c) tokens $\rightarrow$ sec	0.829	0.24
(d) reason-then-sec	0.545	1.08

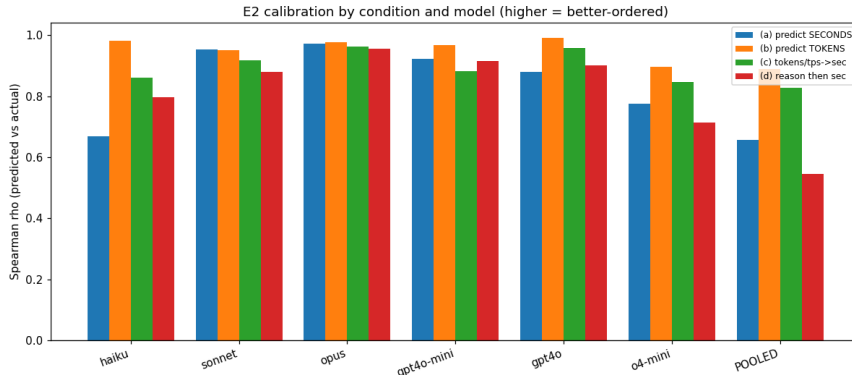


Figure 2: E2: rank correlation of estimate vs. actual, by condition and model. Token space (b) is uniformly high; second space (a) and reasoning (d) are noisier.

5 E3 — Log-gap probe (are agents timestamp-blind?)

Hypothesis. Agents are blind to elapsed time encoded in a transcript: varying only the timestamp gap will not flip a time-sensitive decision.

Setup. 9 scenarios (stock price, weather, inventory, 2FA code, session token, “wait a bit”, geocode, medication dose, food safety), each a short transcript where an action happened at t_0 and a decision is needed at $t_0 + \text{gap}$. The transcript is held *fixed*; only the gap varies over $\{1s, 1min, 1h, 1day, 1week\}$, plus a no-timestamp control. Each scenario declares a human-sensible staleness threshold. One-word forced decision. 5 models, 810 calls. We separate *sensitivity* (does the decision respond to the gap at all?) from *correctness* (does it flip at the right threshold?).

Results (Table 3, Fig. 3). Blindness is *graded*, *not binary*. Sensitivity scales with model strength (opus 1.00 $\rightarrow$ haiku 0.56); when models do attend they are well calibrated (correctness 0.80–0.93). Stripping timestamps (control) changes the decision for no model except opus (33%). The deficit is *attention to the clock*, not temporal judgement.

Verdict. Partially supported, and it refines TicToc’s near-random framing: with a clean single-variable manipulation, strong models *do* respond to gaps and are calibrated; the weak link is inconsistent attention to the clock.

Table 3: E3: sensitivity and correctness by model.

model	sensitivity	correctness
opus	1.00	0.86
sonnet	0.78	0.93
gpt5.2	0.78	0.89
gpt4o	0.67	0.80
haiku	0.56	0.82

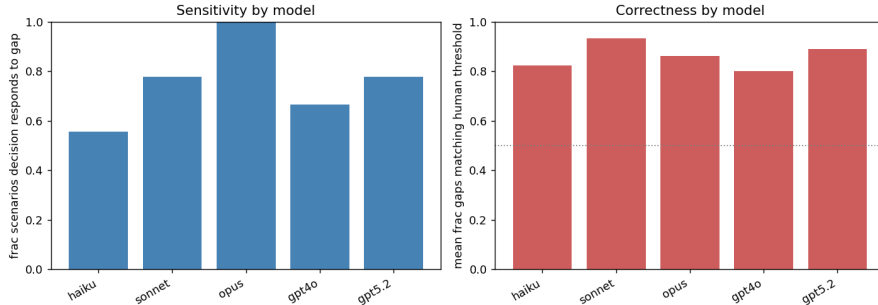


Figure 3: E3: per-scenario sensitivity by model (does the decision change across the gap ladder?). opus reads the clock everywhere; weaker models flatten on extreme-threshold scenarios.

6 E4 — Harness clock (sensor vs text timestamps)

Hypothesis. Much “temporal blindness” is a missing sensor: an explicit harness-injected elapsed-time field beats burying timestamps in text.

Setup. E3-family time-sensitive decisions under three presentation conditions of the *same* scenario+gap: (a) no time signal; (b) absolute timestamps in the transcript text (model must compute the gap); (c) an explicit harness field, e.g. [elapsed since last action: 3 hours 12 minutes]. Gaps straddle each threshold so a correct decision is well defined. 5 models, 1620 calls, 0 failures.

Results (Table 4, Fig. 4). The harness clock wins decisively. Correct-decision rate: 0.494 (none) $\rightarrow$ 0.796 (text) $\rightarrow$ 0.956 (harness). Text over nothing = +0.302; harness over text = +0.159; harness over nothing = +0.461. The lift is largest for weak models and smallest for the reasoning model (o4-mini already computes gaps from text). The cleanest effect is on the fresh side of the threshold, where no-signal models default to “stale” (fresh accuracy 0.23) and the harness clock nearly fixes it (0.94).

Table 4: E4: correct-decision rate by presentation condition (n=540 each).

condition	correct-decision rate
(a) no time signal	0.494
(b) text timestamps	0.796
(c) harness clock	0.956

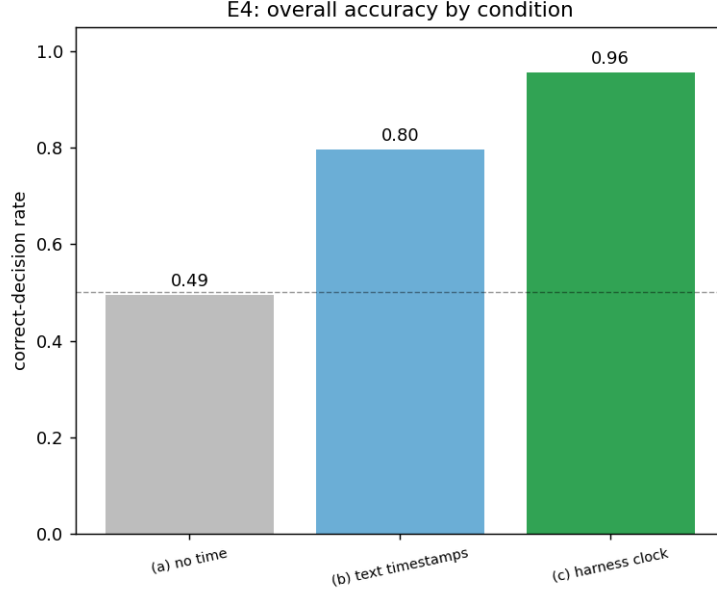


Figure 4: E4: overall correct-decision rate by condition. The sensor (c) closes most of the gap left by text timestamps (b).

Verdict. Supported. A large share of agent temporal blindness is a missing-sensor / tooling problem, addressable at the harness layer without post-training. *Caveat:* the prompt states the staleness rule, so this isolates the *sensor*, not threshold knowledge; the between-condition contrast is the result, not the absolute accuracies (which are optimistic).

7 E5 — Latency-decoupled probe (the linchpin validity check)

Hypothesis. E1/E2’s calibration is a pure output-length proxy: it should survive when latency is output-driven and *collapse* when latency is decoupled from output length.

Setup. Three task families, all logging a PRE seconds-estimate and actual latency: **Type A (reasoning-decoupled)** — short fixed output, varying difficulty (2+2 ... derangement of 8); latency varies via hidden reasoning. **Type B (input-decoupled)** — output pinned to one word, input filler varied 100→30 000 tokens; latency varies via prefill. **Type C (output-driven control)** — output length varies. 5 models, 450 calls.

Results (Table 5, Fig. 5). The prediction holds decisively. Control (C) is strongly calibrated (ρ 0.788, $p < 0.001$), but the reasoning-decoupled (A) and input-decoupled (B) families collapse to no significant correlation *despite* 16× and 9× actual-latency spreads. Even reasoning models fail Type A (gpt5.2 ρ 0.30 n.s., o4-mini −0.36 n.s.). The mechanism is legible in Type B: models emit a near-*constant* time estimate as input grows — e.g. haiku [1.0, 1.0, 1.3, 1.0] seconds for inputs of 100/1k/8k/30k tokens while actual latency rises [0.52, 0.59, 0.71, 0.97]; o4-mini emits [1.0, 1.0, 1.0, 1.0]. A 300× change in input length moves the estimate not at all.

Reconciliation. This resolves E1’s apparent contradiction of the literature. There is no contradiction: in length-dominated tasks models look calibrated; in latency-decoupled tasks (closer to

Table 5: E5: pooled ρ (estimate, actual latency) by task type.

type	pooled ρ	gm(est/act)	actual-latency spread
C — output-driven (control)	0.788	1.30	61.9×
A — reasoning-decoupled	−0.067 (n.s.)	2.01	16.2×
B — input-decoupled	−0.231 (n.s.)	1.44	8.9×

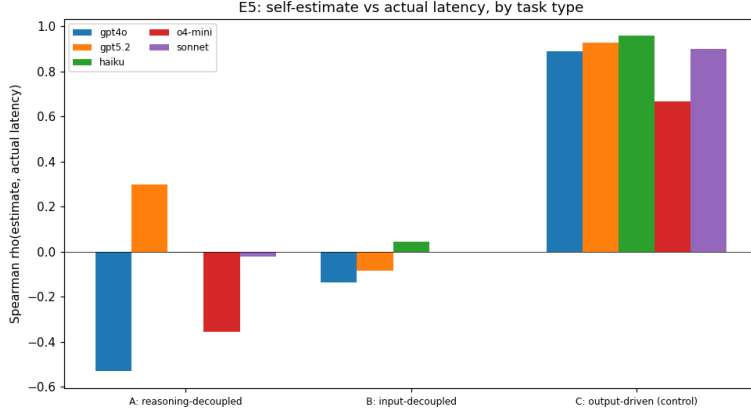


Figure 5: E5: self-estimate vs. actual latency by task type and model. Calibration exists only in the output-driven control (C); it vanishes once latency is decoupled (A, B).

what negative-result papers used) they are blind, exactly as reported. The regime is the hidden variable, and E5 makes it explicit.

Verdict. Confirmed. Self-latency calibration is a pure output-length proxy; any “train-it-in” effort must target the latency the model currently cannot see (reasoning tokens, prefill), not output length.

8 E6 — Length-estimation bias correction

Hypothesis. The $\sim 2\times$ output-length undershoot (E2) closes with in-context interventions, no training.

Setup. 10 length-varying tasks. Per (model $\times$ task $\times$ 3): one generation (actual tokens) and three estimate conditions — **bare** (E2 baseline), **anchors** (few-shot token-count calibration table), **self_revise** (predict, then “you tend to under-count, adjust”, then FINAL: <n>). 6 models, 720 calls.

Results (Table 6, Fig. 6). In-context calibration substantially closes the bias while keeping ranking intact. Pooled gm(pred/actual) moves 0.37 (bare, $\approx 2.7\times$ undershoot) $\rightarrow$ 0.55 (anchors) $\rightarrow$ 0.76 (self-revise), with ρ holding near 0.83. Per-model nuance is the real story: strong models land near 1.0 under self-revision (opus 0.82, sonnet 0.78), weak models *overcorrect* (haiku $\rightarrow$ 2.01, overshoot), and o4-mini stays stuck (0.08 $\rightarrow$ 0.19).

Table 6: E6: pooled length-estimate calibration by condition (1.0 = perfect; <1 = undershoot).

condition	pooled gm(pred/act)	Δ vs bare	pooled ρ
bare	0.37	—	0.842
anchors	0.55	+0.18	0.861
self_revise	0.76	+0.39	0.825

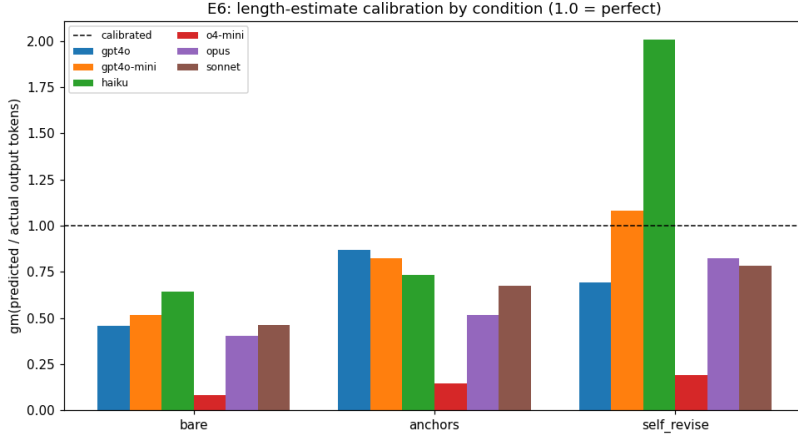


Figure 6: E6: calibration (predicted/actual output tokens) by condition and model; dashed line is perfect calibration. Self-revision pulls strong models to ≈ 1.0 but overshoots haiku and fails to move o4-mini.

Verdict. Largely supported. Target 1 (length self-estimation) is nearly free for non-reasoning models via a fitted in-context correction; fine-tuning effort should be reserved for the reasoning-model hidden-token problem, the one place in-context fixes fail.

9 Synthesis

The six experiments tell one coherent story:

LLMs do not have an internal clock. They have an output-length estimator, and they respond to elapsed time only when it is handed to them explicitly.

Self-perception (E1, E2, E5, E6). Apparent temporal self-awareness is output-length anticipation. It proxies latency whenever latency is length-dominated (E1 ρ up to 0.94; E2 token space ρ 0.89), and collapses the instant latency hides in unseen reasoning or prefill (E5 $\rho \approx 0$, constant estimates). The length undershoot is mostly an in-context calibration error for non-reasoning models (E6 $0.37 \rightarrow 0.76$).

Agency (E3, E4). The same absence of a clock means agents act on elapsed time only when it is explicit and loud: text timestamps get partway (E3/E4 0.80) because strong models can do the arithmetic; an explicit sensor gets almost all the way (E4 0.96). The $0.49 \rightarrow 0.96$ gap is largely a missing-sensor problem, not a missing-capability problem.

The recurring boundary. Across *every* experiment, the reasoning model is the exception: its latency hides in tokens it cannot see, so the length proxy (E1, E2, E5) and every in-context

fix (E6) fail on it. This is the single sharpest, most reproducible finding and the natural research target.

10 Recommendation and next experiments

Two framings are on record and are complementary, not exclusive:

- **“Give it a sensor”** (SUMMARY.md): E4’s harness clock is a cheap, deployable win. Treat it as the *ceiling*.
- **“Train it in”** (direction-train-it-in.md): the absence of a clock is the reason to train the skill in. The research question is how much of the sensor’s gap a model can *internalize* from text alone — internalization fraction = (trained – base)/(ceiling – base).

A note on trainability, corrected from an earlier overstatement: predicting wall-clock seconds is *not* ill-posed within a *fixed* environment. E1’s latency CV of 0.04–0.14 shows the seconds target is reproducible there; the constants (sec/token, overhead) are absorbed by in-environment post-training. Only the *cross-environment* target (from the prompt alone) is ill-posed.

Planned experiments (detail in experiments.md).

- **E7** — harden + scale E3: drop the stated staleness rule, add multi-turn, scale to ~200 scenarios with held-out families. Prerequisite for E8.
- **E8** — internalization: SFT (LoRA) a weak and a strong model on text-timestamp transcripts; report the internalization fraction per model (prediction: highest for weak models). The flagship of the train-it-in direction.
- **E9** — sensor-format ablation: which representation (absolute / elapsed / relative / count-down) carries E4’s lift.
- **E10** — reasoning-model hidden-token probe: can a reasoning model predict its own reasoning length? *Promoted to high priority*: E5 and E6 both isolate this as the one boundary where every proxy and in-context fix fails.

11 Limitations and process notes

Scientific limitations.

- **Small N** (3 trials; 9–12 tasks / scenarios). Trends are clear; tails are thin; no confidence intervals.
- **Regime-specificity** (E1, E2, E6 use output-length-dominated tasks by design; E5 is the explicit decoupling check).
- **Forced one-word decisions** (E3, E4) and integer-second estimates (E1) are lower bounds on capability.
- **Stated thresholds** (E3, E4) isolate the sensor from threshold knowledge; absolute accuracies are optimistic, between-condition contrasts are the result.
- **Cost-estimate dollars** in each report are token-count × approximate prices, not billed amounts; total study spend was well within budget (~\$10 of a \$50+\$50 cap).

Process notes (so the next run avoids these).

- **Concurrent-writer bug.** Orchestration relaunched `run.py` while an earlier instance was still alive; both passed `load_done()` and produced 492 duplicate rows in E4 (the E5/E6 driver hit the same race transiently but dedups every loop). Root cause: `append + load_done()` is not concurrency-safe. Mitigation: single writer per experiment, and `dedup.py` before analysis. A file lock or a single coordinator is the proper fix.
- **Relative interpreter paths.** `.venv/bin/python` breaks inside a `cd-subshell` (exit 127); use the absolute interpreter path in scripts.
- **Background-job lifetime.** Long collections (slow reasoning models) outlived the job runner and were killed; `_driver_e5e6.py` is a self-healing driver that dedups and relaunches any died collection until expected counts are reached.

A File manifest

Path	Contents
<code>research-notes.md</code>	Literature notes and reading list (seed doc)
<code>experiments.md</code>	Full E1–E10 plan and status
<code>SUMMARY.md</code>	Cross-experiment synthesis and recommendation
<code>direction-train-it-in.md</code>	Counterpoint “train-it-in” research direction
<code>common.py</code>	Shared LLM-call helper (roster, timing, retries)
<code>dedup.py</code>	Canonicalize <code>results.jsonl</code> by logical key
<code>_driver_e5e6.py</code>	Self-healing overnight collection driver
<code>requirements.txt</code>	Pinned Python dependencies
<code>eN-*/run.py</code>	Data collection (resumable)
<code>eN-*/analyze.py</code>	Statistics + figures from <code>results.jsonl</code>
<code>eN-*/results.jsonl</code>	Raw per-call data (canonical, deduped)
<code>eN-*/REPORT.md</code>	Self-contained per-experiment findings
<code>eN-*/*.png</code>	Figures
<code>e4-harness-clock/NOTES.md</code>	Debugging log (opus temperature, dedup)

B Experiment summary table

Exp	Calls	One-line result
E1	648	Self-latency calibration is a length proxy (ρ up to 0.94); not 4–7 $\times$ overshoot.
E2	864	Token space beats seconds (ρ 0.89 vs 0.66); $\sim 2\times$ length undershoot.
E3	810	Timestamp-sensitivity graded by model strength (opus 1.00 $\rightarrow$ haiku 0.56).
E4	1620	Harness clock 0.96 > text 0.80 > none 0.49.
E5	450	Calibration collapses when latency is decoupled ($\rho \approx 0$); confirms length proxy.
E6	720	Self-revision closes the undershoot 0.37 $\rightarrow$ 0.76 (non-reasoning models).